load_trait_records_from_vegvault.R

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Contents

`load_trait_records_from_vegvault` *Load Trait Records from VegVault*

Description

Loads functional trait records from the VegVault SQLite database for specified trait domains. Optional geographic filtering limits records to datasets within the specified longitude/latitude bounding box. Taxon names are returned as-is (‘classify_to = NULL‘) for downstream manual classification.

Usage

```
load_trait_records_from_vegvault(
  path_vegvault = here::here("Data/Input/VegVault.sqlite"),
  vec_trait_domain_names = NULL,
  vec_longitude_limits = NULL,
  vec_latitude_limits = NULL
)
```

Arguments

`path_vegvault` A character string specifying the path to the VegVault SQLite database (default: ‘here::here("Data/Input/VegVault.sqlite")‘).

`vec_trait_domain_names`
A character vector specifying the trait domain names to load (e.g. ‘c("Specific leaf area (SLA)", "Plant height vegetative")‘).

`vec_longitude_limits`
A numeric vector of length 2 giving the longitude limits ‘c(min, max)‘ for geographic filtering. Must be supplied together with ‘vec_latitude_limits‘. If ‘NULL‘ (default) no geographic filter is applied.

`vec_latitude_limits`
A numeric vector of length 2 giving the latitude limits ‘c(min, max)‘ for geographic filtering. Must be supplied together with ‘vec_longitude_limits‘. If ‘NULL‘ (default) no geographic filter is applied.

Details

The function performs the following steps:

1. Validates input parameters. 2. Checks the presence of the VegVault SQLite database. 3. Builds the vaultkeepr query plan (lazy SQL). If geographic limits are supplied, datasets are filtered to the bounding box using ‘vaultkeepr::select_dataset_by_geo()‘ before sample retrieval. If vaultkeepr raises an error during plan assembly (e.g. no data for the given domain names), the error is caught and re-thrown via ‘cli::cli_abort()‘ with the original message preserved. 4. Retrieves trait values using ‘classify_to = NULL‘ to preserve raw taxon names for downstream manual classification via the same taxospace + auxiliary-table pipeline used for community data. 5. Returns the loaded data as a flat data frame.

Value

A flat data frame containing loaded trait records with columns including ‘taxon_name‘ (original, unclassified), ‘trait_domain_name‘, ‘trait_name‘, and ‘trait_value‘.

See Also

[extract_data_from_vegvault()]

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